COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:		**SECTION:**	
DATE PERFORMED:		**DATE SUBMITTED:**	
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

Step 1 - assign machines

Machine	Username	IP address	Purpose
Workstation (Ubuntu)	wdjavillonar	192.168.56.103	Runs Ansible
Server 1 (Ubuntu)	serv1javillonar	192.168.56.101	Elasticsearch
Server 2 (Ubuntu)	serv2javillonar	192.168.56.102	Kibana
Server 3 (Ubuntu)	javillonar	192.168.56.104	Logstash
CentOS	cjavillonar	192.168.1.23	Additional installation and verification

Step 2 — Prepare the Ubuntu workstation

2.1 Update Ubuntu and install the required tools

```
wdjavillonar@workstation:~$ sudo apt update
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,144 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,221 kB]
Fetched 2,491 kB in 5s (535 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
165 packages can be upgraded. Run 'apt list --upgradable' to see them.
wdjavillonar@workstation:~$ sudo apt install -y ansible git openssh-client openssh-server cur
 wget gnupg ca-certificates apt-transport-https
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ansible is already the newest version (9.2.0+dfsg-0ubuntu5).
git is already the newest version (1:2.43.0-1ubuntu7.3).
curl is already the newest version (8.5.0-2ubuntu10.15).
wget is already the newest version (1.21.4-1ubuntu4.5).
```

2.2 Verify the installation

```
wdjavillonar@workstation:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/wdjavillonar/.ansible/plugins/modules', '/usr/share/
ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/wdjavillonar/.ansible/collections:/usr/share/ansible/col
lections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jul 15 2026, 23:46:41) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
wdjavillonar@workstation:~$ git --version
git version 2.43.0
wdjavillonar@workstation:~$ ssh -V
OpenSSH_9.6p1 Ubuntu-3ubuntu13.20, OpenSSL 3.0.13 30 Jan 2024
```

2.3 Create a fresh project folder

```
wdjavillonar@workstation:~$ mkdir -p ~/HOA10.1_Elastic_Stack
wdjavillonar@workstation:~$ cd ~/HOA10.1_Elastic_Stack
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ pwd
/home/wdjavillonar/HOA10.1_Elastic_Stack
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$
```

2.4 Check SSH connectivity to your three Ubuntu servers

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ssh serv1javillonar@192.168.56.101
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

103 updates can be applied immediately.
59 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Oct  6 21:05:18 2026 from 192.168.56.103
serv1javillonar@serv1javillonar-VirtualBox:~$ exit
logout
Connection to 192.168.56.101 closed.
```

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ssh serv2javillonar@192.168.56.102
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

59 updates can be applied immediately.
23 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Oct  6 21:05:18 2026 from 192.168.56.103
serv2javillonar@server2:~$ exit
logout
Connection to 192.168.56.102 closed.
```

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ssh javillonar@192.168.56.104
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

69 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue Oct  6 13:07:14 2026 from 192.168.56.103
javillonar@serv3javi:~$ exit
logout
Connection to 192.168.56.104 closed.
```

Step 3 — Create your Ansible project structure
3.1 Open project folder
3.2 Create an SSH key

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ cd ~/HOA10.1_Elastic_Stack
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ssh-keygen -t ed25519 -C "hoa10.1-ansible"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/wdjavillonar/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/wdjavillonar/.ssh/id_ed25519
Your public key has been saved in /home/wdjavillonar/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:kg2zhxpiMoNGmb1k/cZw1Kumx9GN6c9XKeaH8XnZnzU hoa10.1-ansible
The key's randomart image is:
+--[ED25519 256]--+
|       ..        |
|  + . .  .       |
| + + oo.  .      |
|o o . =B .       |
|=.o.. =+S +     .|
|.= . o.* + . + o |
|    . + o   o *E=|
|     . o ..  + =*|
|      .   .o. ..+|
+----[SHA256]-----+
```

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ssh-copy-id serv1javillonar@192.168.56.101
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that a
re already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is t
o install the new keys

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'serv1javillonar@192.168.56.101'"
and check to make sure that only the key(s) you wanted were added.

wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ssh-copy-id serv2javillonar@192.168.56.102
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that a
re already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is t
o install the new keys

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'serv2javillonar@192.168.56.102'"
and check to make sure that only the key(s) you wanted were added.

wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ssh-copy-id javillonar@192.168.56.104
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that a
re already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is t
o install the new keys

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'javillonar@192.168.56.104'"
and check to make sure that only the key(s) you wanted were added.
```

Step 4 — Create the Ansible inventory

4.1 Create the inventory file

```
  GNU nano 7.2                                    inventory.ini *
[elasticsearch]
serv1 ansible_host=192.168.56.101 ansible_user=serv1javillonar

[kibana]
serv2 ansible_host=192.168.56.102 ansible_user=serv2javillonar

[logstash]
serv3 ansible_host=192.168.56.104 ansible_user=javillonar

[ubuntu:children]
elasticsearch
kibana
logstash

[centos]
cent1 ansible_host=192.168.1.23 ansible_user=cjavillonar
```

4.2 Check your inventory

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-inventory -i inventory.ini --list
{
    "_meta": {
        "hostvars": {
            "cent1": {
                "ansible_host": "192.168.1.23",
                "ansible_user": "cjavillonar"
            },
            "serv1": {
                "ansible_host": "192.168.56.101",
                "ansible_user": "serv1javillonar"
            },
            "serv2": {
                "ansible_host": "192.168.56.102",
                "ansible_user": "serv2javillonar"
            },
            "serv3": {
                "ansible_host": "192.168.56.104",
                "ansible_user": "javillonar"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
            "ubuntu",
            "centos"
        ]
    },
    "centos": {
        "hosts": [
            "cent1"
        ]
    },
```

```
        }
    }
},
"all": {
    "children": [
        "ungrouped",
        "ubuntu",
        "centos"
    ]
},
"centos": {
    "hosts": [
        "cent1"
    ]
},
"elasticsearch": {
    "hosts": [
        "serv1"
    ]
},
"kibana": {
    "hosts": [
        "serv2"
    ]
},
"logstash": {
    "hosts": [
        "serv3"
    ]
},
"ubuntu": {
    "children": [
        "elasticsearch",
        "kibana",
        "logstash"
```

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible ubuntu -i inventory.ini -m ping
serv1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
serv3 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
serv2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Step 5 — Create the three Ansible roles

5.1 Generate the roles

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-galaxy role init roles/elasticsearch
- Role roles/elasticsearch was created successfully
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-galaxy role init roles/kibana
- Role roles/kibana was created successfully
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-galaxy role init roles/logstash
- Role roles/logstash was created successfully
```

5.2 Check the structure

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ find roles -maxdepth 3 -type f | sort
roles/elasticsearch/defaults/main.yml
roles/elasticsearch/handlers/main.yml
roles/elasticsearch/meta/main.yml
roles/elasticsearch/README.md
roles/elasticsearch/tasks/main.yml
roles/elasticsearch/tests/inventory
roles/elasticsearch/tests/test.yml
roles/elasticsearch/vars/main.yml
roles/kibana/defaults/main.yml
roles/kibana/handlers/main.yml
roles/kibana/meta/main.yml
roles/kibana/README.md
roles/kibana/tasks/main.yml
roles/kibana/tests/inventory
roles/kibana/tests/test.yml
roles/kibana/vars/main.yml
roles/logstash/defaults/main.yml
roles/logstash/handlers/main.yml
roles/logstash/meta/main.yml
roles/logstash/README.md
roles/logstash/tasks/main.yml
roles/logstash/tests/inventory
roles/logstash/tests/test.yml
roles/logstash/vars/main.yml
```

Step 6 — Configure administrative privileges

7.1 Check your project directory

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ls roles/elasticsearch
defaults  files  handlers  meta  README.md  tasks  templates  tests  vars
```

7.2 Create the Elasticsearch configuration template

```
  GNU nano 7.2          roles/elasticsearch/templates/elasticsearch.yml.j2 *
cluster.name: elastic-lab
node.name: "{{ inventory_hostname }}"
network.host: 0.0.0.0
http.port: 9200
discovery.type: single-node
xpack.security.enabled: false
```

7.3 Create the Elasticsearch installation tasks

GNU nano 7.2 roles/elasticsearch/tasks/main.yml *

```
---
# tasks file for roles/elasticsearch
- name: Install required packages
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - gnupg
      - curl
    state: present
    update_cache: true
  become: true

- name: Create directory for Elastic signing key
  ansible.builtin.file:
    path: /usr/share/keyrings
    state: directory
    mode: '0755'
  become: true

- name: Download Elastic signing key
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /usr/share/keyrings/elasticsearch-keyring.asc
    mode: '0644'
  become: true

- name: Add Elastic 8.x repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.asc] https://artifacts.el
    state: present
    filename: elastic-8.x
    update_cache: true
  become: true
```

```
  GNU nano 7.2                    roles/elasticsearch/tasks/main.yml *
  become: true

- name: Add Elastic 8.x repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.asc] https://artifacts.el
    state: present
    filename: elastic-8.x
    update_cache: true
  become: true

- name: Install Elasticsearch
  ansible.builtin.apt:
    name: elasticsearch
    state: present
    update_cache: true
  become: true

- name: Configure Elasticsearch
  ansible.builtin.template:
    src: elasticsearch.yml.j2
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'
  become: true
  notify: Restart Elasticsearch

- name: Enable and start Elasticsearch
  ansible.builtin.systemd:
    name: elasticsearch
    enabled: true
    state: started
  become: true
```

7.4 Create the Elasticsearch handler

```
  GNU nano 7.2                    roles/elasticsearch/handlers/main.yml
---
# handlers file for roles/elasticsearch
- name: Restart Elasticsearch
  ansible.builtin.systemd:
    name: elasticsearch
    state: restarted
    enabled: true
  become: true
```

Step 8 — Create the main Ansible playbook

8.1 Create site.yml

```
  GNU nano 7.2                                          site.yml *
---
- name: Install and configure Elasticsearch
  hosts: elasticsearch
  become: true
  roles:
    - elasticsearch
```

8.2 Check the playbook syntax

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-playbook -i inventory.ini site.yml
-syntax-check

playbook: site.yml
```

8.3 Run the Elasticsearch installation

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-playbook -i inventory.ini site.yml -
-ask-become-pass
BECOME password:

PLAY [Install and configure Elasticsearch] *************************************************

TASK [Gathering Facts] *********************************************************************
ok: [serv1]

TASK [elasticsearch : Install required packages] *******************************************
ok: [serv1]

TASK [elasticsearch : Create directory for Elastic signing key] ****************************
ok: [serv1]

TASK [elasticsearch : Download Elastic signing key] ****************************************
ok: [serv1]

TASK [elasticsearch : Add Elastic 8.x repository] ******************************************
ok: [serv1]

TASK [elasticsearch : Install Elasticsearch] ***********************************************
ok: [serv1]

TASK [elasticsearch : Configure Elasticsearch] *********************************************
changed: [serv1]

TASK [elasticsearch : Enable and start Elasticsearch] **************************************
changed: [serv1]

RUNNING HANDLER [elasticsearch : Restart Elasticsearch] ************************************
changed: [serv1]

PLAY RECAP *********************************************************************************
serv1                      : ok=9    changed=3    unreachable=0    failed=0    skipped=0    re
scued=0    ignored=0
```

8.4 Verify Elasticsearch

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible elasticsearch -i inventory.ini -b -m
 command -a "systemctl is-active elasticsearch"
serv1 | CHANGED | rc=0 >>
active
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible elasticsearch -i inventory.ini -b -m
 command -a "dpkg-query -W -f='${Version}\n' elasticsearch"
serv1 | CHANGED | rc=0 >>
```

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible elasticsearch -i inventory.ini -m ur
i -a "url=http://192.168.56.101:9200 method=GET"
serv1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "content_length": "529",
    "content_type": "application/json",
    "cookies": {},
    "cookies_string": "",
    "elapsed": 0,
    "json": {
        "cluster_name": "elastic-lab",
        "cluster_uuid": "A31zWD8-THiSEJl-c2h7Fw",
        "name": "serv1",
        "tagline": "You Know, for Search",
        "version": {
            "build_date": "2026-09-30T23:11:55.493565832Z",
            "build_flavor": "default",
            "build_hash": "6940e499f8a38841809b8e295a20c6421c95d9e4",
            "build_snapshot": false,
            "build_type": "deb",
            "lucene_version": "9.12.2",
            "minimum_index_compatibility_version": "7.0.0",
            "minimum_wire_compatibility_version": "7.17.0",
            "number": "8.19.23"
        }
    },
    "msg": "OK (529 bytes)",
    "redirected": false,
    "status": 200,
    "url": "http://192.168.56.101:9200",
    "x_elastic_product": "Elasticsearch"
}
```

Step 9 — Create the Kibana configuration template

```
  GNU nano 7.2                    roles/kibana/templates/kibana.yml.j2 *
server.port: 5601
server.host: "0.0.0.0"
server.name: "{{ inventory_hostname }}"

elasticsearch.hosts: ["http://192.168.56.101:9200"]
```

Step 10 — Create the Kibana installation tasks

```
  GNU nano 7.2                        roles/kibana/tasks/main.yml *
---
# tasks file for roles/kibana
- name: Install required packages
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - gnupg
      - curl
  state: present
    update_cache: true
  become: true

- name: Add Elastic signing key
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /usr/share/keyrings/elastic.asc
    mode: '0644'
  become: true

- name: Add Elastic 8.x repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elastic.asc] https://artifacts.elastic.co/packa
    state: present
    filename: elastic-8.x
    update_cache: true
  become: true

- name: Install Kibana
  ansible.builtin.apt:
    name: kibana
    state: present
    update_cache: true
  become: true
```

```
- name: Configure Kibana
  ansible.builtin.template:
    src: kibana.yml.j2
    dest: /etc/kibana/kibana.yml
    owner: root
    group: kibana
    mode: '0660'
  become: true
  notify: Restart Kibana

- name: Enable and start Kibana
  ansible.builtin.systemd:
    name: kibana
    enabled: true
    state: started
  become: true
```

Step 11 — Create the Kibana handler

```
  GNU nano 7.2                          roles/kibana/handlers/main.yml *
---
# handlers file for roles/kibana
- name: Restart Kibana
  ansible.builtin.systemd:
    name: kibana
    state: restarted
    enabled: true
  become: true
```

Step 12 — Add Kibana to the main playbook

```
  GNU nano 7.2                                    site.yml *
---
- name: Install and configure Elasticsearch
  hosts: elasticsearch
  become: true
  roles:
    - elasticsearch

- name: Install and configure Kibana
  hosts: kibana
  become: true
  roles:
    - kibana
```

Then check

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-playbook -i inventory.ini site.yml -
-syntax-check

playbook: site.yml
```

Step 13 — Run the playbook

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-playbook -i inventory.ini site.yml -
--ask-become-pass
BECOME password:

PLAY [Install and configure Elasticsearch] *************************************************

TASK [Gathering Facts] *********************************************************************
ok: [serv1]

TASK [elasticsearch : Install required packages] *******************************************
ok: [serv1]

TASK [elasticsearch : Create directory for Elastic signing key] ****************************
ok: [serv1]

TASK [elasticsearch : Download Elastic signing key] ****************************************
ok: [serv1]

TASK [elasticsearch : Add Elastic 8.x repository] ******************************************
ok: [serv1]

TASK [elasticsearch : Install Elasticsearch] ***********************************************
ok: [serv1]

TASK [elasticsearch : Configure Elasticsearch] *********************************************
ok: [serv1]

TASK [elasticsearch : Enable and start Elasticsearch] **************************************
ok: [serv1]

PLAY [Install and configure Kibana] ********************************************************

TASK [Gathering Facts] *********************************************************************
ok: [serv2]
```

```
TASK [kibana : Install required packages] ***************************************
ok: [serv2]

TASK [kibana : Add Elastic signing key] *****************************************
ok: [serv2]

TASK [kibana : Add Elastic 8.x repository] **************************************
ok: [serv2]

TASK [kibana : Install Kibana] **************************************************
ok: [serv2]

TASK [kibana : Configure Kibana] ************************************************
changed: [serv2]

TASK [kibana : Enable and start Kibana] *****************************************
ok: [serv2]

RUNNING HANDLER [kibana : Restart Kibana] ***************************************
changed: [serv2]

PLAY RECAP **********************************************************************
serv1                      : ok=8    changed=0    unreachable=0    failed=0    skipped=0    re
scued=0    ignored=0
serv2                      : ok=8    changed=2    unreachable=0    failed=0    skipped=0    re
scued=0    ignored=0
```

Step 14— Verify Kibana is running

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible kibana -i inventory.ini -b -m comman
d -a "systemctl is-active kibana" --ask-become-pass
BECOME password:
serv2 | CHANGED | rc=0 >>
active
```

Step 15 — Test Kibana's web service

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible kibana -i inventory.ini -m uri -a "u
rl=http://192.168.56.102:5601/api/status method=GET status_code=200,302,503"
serv2 | SUCCESS => {
    "accept_ranges": "bytes",
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_control": "private, no-cache, no-store, must-revalidate",
    "changed": false,
    "connection": "close",
    "content_length": "20450",
    "content_security_policy": "script-src 'report-sample' 'self'; worker-src 'report-sample'
'self' blob:; style-src 'report-sample' 'self' 'unsafe-inline'",
    "content_security_policy_report_only": "form-action 'report-sample' 'self'; default-src 'r
eport-sample' 'none'; font-src 'report-sample' 'self'; img-src 'report-sample' 'self' data: ti
les.maps.elastic.co; connect-src 'report-sample' 'self' telemetry.elastic.co telemetry-staging
.elastic.co feeds.elastic.co tiles.maps.elastic.co vector.maps.elastic.co",
    "content_type": "application/json; charset=utf-8",
    "cookies": {},
    "cookies_string": "",
    "cross_origin_opener_policy": "same-origin",
    "date": "Wed, 14 Oct 2026 14:32:01 GMT",
    "elapsed": 0,
    "elastic_api_version": "2023-10-31",
    "json": {
        "metrics": {
            "collection_interval_in_millis": 5000,
            "concurrent_connections": 0,
            "elasticsearch_client": {
                "totalActiveSockets": 1,
                "totalIdleSockets": 2,
                "totalQueuedRequests": 0
            },
            "last_updated": "2026-10-14T14:32:01.344Z",
            "os": {
                "distro": "Ubuntu",
                "distroRelease": "Ubuntu-24.04",
```

```
        "concurrent_connections": 0,
        "elasticsearch_client": {
            "totalActiveSockets": 1,
            "totalIdleSockets": 2,
            "totalQueuedRequests": 0
        },
        "last_updated": "2026-10-14T14:32:01.344Z",
        "os": {
            "distro": "Ubuntu",
            "distroRelease": "Ubuntu-24.04",
            "load": {
                "15m": 0.18,
                "1m": 0.38,
                "5m": 0.25
            },
            "memory": {
                "free_in_bytes": 2250727424,
                "total_in_bytes": 4101586944,
                "used_in_bytes": 1850859520
            },
            "platform": "linux",
            "platformRelease": "linux-7.0.0-38-generic",
            "uptime_in_millis": 3689160
        },
        "process": {
            "event_loop_delay": 41.615359,
            "event_loop_delay_histogram": {
                "exceeds": 0,
                "fromTimestamp": "2026-10-14T14:31:56.343Z",
                "lastUpdatedAt": "2026-10-14T14:32:01.343Z",
                "max": 41.615359,
                "mean": 13.1024896,
                "min": 9.388032,
                "percentiles": {
                    "50": 12.533759,
                    "75": 14.409727,
                    "95": 17.088511,
                    "99": 19.267583
```

```
            "utilization": 0.01033430928625802
        },
        "memory": {
            "array_buffers_in_bytes": 1472994,
            "external_in_bytes": 5894417,
            "heap": {
                "size_limit": 2252341248,
                "total_in_bytes": 504238080,
                "used_in_bytes": 424417688
            },
            "resident_set_size_in_bytes": 687960064
        },
        "pid": 10897,
        "uptime_in_millis": 288242.64049900003
    },
    "processes": [
        {
            "event_loop_delay": 41.615359,
            "event_loop_delay_histogram": {
                "exceeds": 0,
                "fromTimestamp": "2026-10-14T14:31:56.343Z",
                "lastUpdatedAt": "2026-10-14T14:32:01.343Z",
                "max": 41.615359,
                "mean": 13.1024896,
                "min": 9.388032,
                "percentiles": {
                    "50": 12.533759,
                    "75": 14.409727,
                    "95": 17.088511,
                    "99": 19.267583
                },
                "stddev": 2.803289640132146
            },
            "event_loop_utilization": {
                "active": 51.67445000016596,
                "idle": 4948.606513999985,
                "utilization": 0.01033430928625802
            },
```

```
            "total": 0
        },
        "response_times": {
            "avg_in_millis": 0,
            "max_in_millis": 0
        }
    },
    "name": "serv2",
    "status": {
        "core": {
            "elasticsearch": {
                "level": "available",
                "meta": {
                    "incompatibleNodes": [],
                    "warningNodes": []
                },
                "summary": "Elasticsearch is available"
            },
            "savedObjects": {
                "level": "available",
                "meta": {
                    "migratedIndices": {
                        "migrated": 0,
                        "patched": 7,
                        "skipped": 0
                    }
                },
                "summary": "SavedObjects service has completed migrations and is available
"
            }
        },
        "overall": {
            "level": "available",
            "summary": "All services and plugins are available"
        },
        "plugins": {
            "actions": {
                "level": "available",
```

```
    "aiAssistantManagementSelection": {
        "level": "available",
        "summary": "All services and plugins are available"
    },
    "aiops": {
        "level": "available",
        "summary": "All services and plugins are available"
    },
    "alerting": {
        "level": "available",
        "reported": true,
        "summary": "Alerting is (probably) ready"
    },
    "apm": {
        "level": "available",
        "summary": "All services and plugins are available"
    },
    "apmDataAccess": {
        "level": "available",
        "summary": "All services and plugins are available"
    },
    "apmSourcesAccess": {
        "level": "available",
        "summary": "All services and plugins are available"
    },
    "banners": {
        "level": "available",
        "summary": "All services and plugins are available"
    },
    "bfetch": {
        "level": "available",
        "summary": "All services are available"
    },
    "canvas": {
        "level": "available",
        "summary": "All services and plugins are available"
    },
    "cases": {
```



Step 16 — Create the Logstash configuration template

```
  GNU nano 7.2                  roles/logstash/templates/logstash.conf.j2 *
input {
  beats {
    port => 5044
  }
}

filter {
}

output {
  elasticsearch {
    hosts => ["http://192.168.56.101:9200"]
    index => "logstash-%{+YYYY.MM.dd}"
  }

  stdout {
    codec => rubydebug
  }
}
```

Step 17 — Create the Logstash installation tasks

GNU nano 7.2 roles/logstash/tasks/main.yml *

```yaml
---
# tasks file for roles/logstash
- name: Install required packages
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - gnupg
      - curl
    state: present
    update_cache: true
  become: true

- name: Add Elastic signing key
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /usr/share/keyrings/elastic.asc
    mode: '0644'
  become: true

- name: Add Elastic 8.x repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elastic.asc] https://artifacts.elastic.co/packa>
    state: present
    filename: elastic-8.x
    update_cache: true
  become: true

- name: Install Logstash
  ansible.builtin.apt:
    name: logstash
    state: present
    update_cache: true
  become: true
```

```yaml
- name: Configure Logstash pipeline
  ansible.builtin.template:
    src: logstash.conf.j2
    dest: /etc/logstash/conf.d/logstash.conf
    owner: root
    group: root
    mode: '0644'
  become: true
  notify: Restart Logstash

- name: Enable and start Logstash
  ansible.builtin.systemd:
    name: logstash
    enabled: true
    state: started
  become: true
```

Step 18 — Create the Logstash handler
Step 19 — Update the main playbook

```
  GNU nano 7.2                          roles/logstash/handlers/main.yml *
---
# handlers file for roles/logstash
- name: Restart Logstash
  ansible.builtin.systemd:
    name: logstash
    state: restarted
    enabled: true
  become: true
```

Step20 — Update the main playbook

```
  GNU nano 7.2                                    site.yml *
---
- name: Install and configure Elasticsearch
  hosts: elasticsearch
  become: true
  roles:
    - elasticsearch

- name: Install and configure Kibana
  hosts: kibana
  become: true
  roles:
    - kibana

- name: Install and configure Logstash
  hosts: logstash
  become: true
  roles:
    - logstash
```

Step 21— Check the syntax

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-playbook -i inventory.ini site.yml
-syntax-check

playbook: site.yml
```

Step 22 — Install Logstash

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible-playbook -i inventory.ini site.yml --ask-become-pass
BECOME password:

PLAY [Install and configure Elasticsearch] *************************************

TASK [Gathering Facts] *********************************************************
ok: [serv1]

TASK [elasticsearch : Install required packages] *******************************
ok: [serv1]

TASK [elasticsearch : Create directory for Elastic signing key] ****************
ok: [serv1]

TASK [elasticsearch : Download Elastic signing key] ****************************
ok: [serv1]

TASK [elasticsearch : Add Elastic 8.x repository] ******************************
ok: [serv1]

TASK [elasticsearch : Install Elasticsearch] ***********************************
ok: [serv1]

TASK [elasticsearch : Configure Elasticsearch] *********************************
ok: [serv1]

TASK [elasticsearch : Enable and start Elasticsearch] **************************
ok: [serv1]

PLAY [Install and configure Kibana] ********************************************

TASK [Gathering Facts] *********************************************************
ok: [serv2]
```

```
TASK [kibana : Enable and start Kibana] ****************************************************
ok: [serv2]

PLAY [Install and configure Logstash] ******************************************************

TASK [Gathering Facts] *********************************************************************
ok: [serv3]

TASK [logstash : Install required packages] ************************************************
ok: [serv3]

TASK [logstash : Add Elastic signing key] **************************************************
ok: [serv3]

TASK [logstash : Add Elastic 8.x repository] ***********************************************
ok: [serv3]

TASK [logstash : Install Logstash] *********************************************************
ok: [serv3]

TASK [logstash : Configure Logstash pipeline] **********************************************
changed: [serv3]

TASK [logstash : Enable and start Logstash] ************************************************
changed: [serv3]

RUNNING HANDLER [logstash : Restart Logstash] **********************************************
changed: [serv3]

PLAY RECAP *********************************************************************************
serv1                      : ok=8    changed=0    unreachable=0    failed=0    skipped=0    re
scued=0    ignored=0
serv2                      : ok=7    changed=0    unreachable=0    failed=0    skipped=0    re
scued=0    ignored=0
serv3                      : ok=8    changed=3    unreachable=0    failed=0    skipped=0    re
scued=0    ignored=0
```

Step 23 — Verify Logstash is running

Step 24 — Check the Logstash version

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible logstash -i inventory.ini -b -m comm
and -a "systemctl is-active logstash" --ask-become-pass
BECOME password:
serv3 | CHANGED | rc=0 >>
active
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible logstash -i inventory.ini -b -m comm
and -a "/usr/share/logstash/bin/logstash --version" --ask-become-pass
BECOME password:
serv3 | CHANGED | rc=0 >>
logstash 8.19.23Using bundled JDK: /usr/share/logstash/jdk
```

Step 25— Check the listening port

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible logstash -i inventory.ini -b -m shel
l -a "ss -lntp | grep ':5044'" --ask-become-pass
BECOME password:
serv3 | CHANGED | rc=0 >>
LISTEN 0      4096                   *:5044             *:*    users:(("java",pid=10457,fd=106
))
```

Step 26 — Verify all three services

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible elasticsearch -i inventory.ini -b -m
 command -a "systemctl is-active elasticsearch" --ask-become-pass
BECOME password:
serv1 | CHANGED | rc=0 >>
active
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible kibana -i inventory.ini -b -m comman
d -a "systemctl is-active kibana" --ask-become-pass
BECOME password:
serv2 | CHANGED | rc=0 >>
active
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible logstash -i inventory.ini -b -m comm
and -a "systemctl is-active logstash" --ask-become-pass
BECOME password:
serv3 | CHANGED | rc=0 >>
active
```

Step 27— Verify Elasticsearch and Kibana APIs

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible elasticsearch -i inventory.ini -m u
i -a "url=http://192.168.56.101:9200 method=GET"
serv1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "content_length": "529",
    "content_type": "application/json",
    "cookies": {},
    "cookies_string": "",
    "elapsed": 0,
    "json": {
        "cluster_name": "elastic-lab",
        "cluster_uuid": "A31zWD8-THiSEJl-c2h7Fw",
        "name": "serv1",
        "tagline": "You Know, for Search",
        "version": {
            "build_date": "2026-09-30T23:11:55.493565832Z",
            "build_flavor": "default",
            "build_hash": "6940e499f8a38841809b8e295a20c6421c95d9e4",
            "build_snapshot": false,
            "build_type": "deb",
            "lucene_version": "9.12.2",
            "minimum_index_compatibility_version": "7.0.0",
            "minimum_wire_compatibility_version": "7.17.0",
            "number": "8.19.23"
        }
    },
    "msg": "OK (529 bytes)",
    "redirected": false,
    "status": 200,
    "url": "http://192.168.56.101:9200",
    "x_elastic_product": "Elasticsearch"
}
```

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ ansible kibana -i inventory.ini -m uri -a "u
rl=http://192.168.56.102:5601/api/status method=GET status_code=200"
serv2 | SUCCESS => {
    "accept_ranges": "bytes",
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "cache_control": "private, no-cache, no-store, must-revalidate",
    "changed": false,
    "connection": "close",
    "content_length": "20457",
    "content_security_policy": "script-src 'report-sample' 'self'; worker-src 'report-sample'
'self' blob:; style-src 'report-sample' 'self' 'unsafe-inline'",
    "content_security_policy_report_only": "form-action 'report-sample' 'self'; default-src 'r
eport-sample' 'none'; font-src 'report-sample' 'self'; img-src 'report-sample' 'self' data: ti
les.maps.elastic.co; connect-src 'report-sample' 'self' telemetry.elastic.co telemetry-staging
.elastic.co feeds.elastic.co tiles.maps.elastic.co vector.maps.elastic.co",
    "content_type": "application/json; charset=utf-8",
    "cookies": {},
    "cookies_string": "",
    "cross_origin_opener_policy": "same-origin",
    "date": "Wed, 14 Oct 2026 14:46:05 GMT",
    "elapsed": 0,
    "elastic_api_version": "2023-10-31",
    "json": {
        "metrics": {
            "collection_interval_in_millis": 5000,
            "concurrent_connections": 0,
            "elasticsearch_client": {
                "totalActiveSockets": 0,
                "totalIdleSockets": 3,
                "totalQueuedRequests": 0
            },
            "last_updated": "2026-10-14T14:46:02.774Z",
            "os": {
```

Final check:

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ find . -maxdepth 4 -type f | sort
./inventory.ini
./roles/elasticsearch/defaults/main.yml
./roles/elasticsearch/handlers/main.yml
./roles/elasticsearch/meta/main.yml
./roles/elasticsearch/README.md
./roles/elasticsearch/tasks/main.yml
./roles/elasticsearch/templates/elasticsearch.yml.j2
./roles/elasticsearch/tests/inventory
./roles/elasticsearch/tests/test.yml
./roles/elasticsearch/vars/main.yml
./roles/kibana/defaults/main.yml
./roles/kibana/handlers/main.yml
./roles/kibana/meta/main.yml
./roles/kibana/README.md
./roles/kibana/tasks/main.yml
./roles/kibana/templates/kibana.yml.j2
./roles/kibana/tests/inventory
./roles/kibana/tests/test.yml
./roles/kibana/vars/main.yml
./roles/logstash/defaults/main.yml
./roles/logstash/handlers/main.yml
./roles/logstash/meta/main.yml
./roles/logstash/README.md
./roles/logstash/tasks/main.yml
./roles/logstash/templates/logstash.conf.j2
./roles/logstash/tests/inventory
./roles/logstash/tests/test.yml
./roles/logstash/vars/main.yml
./site.yml
```

GitHub

```
wdjavillonar@workstation:~/HOA10.1_Elastic_Stack$ cd ~
ls -d CPE212_Javillonar
CPE212_Javillonar
wdjavillonar@workstation:~$ cp -r ~/HOA10.1_Elastic_Stack ~/CPE212_Javillonar/HOA10.1_Log_Mon
toring_New
wdjavillonar@workstation:~$ cd ~/CPE212_Javillonar
wdjavillonar@workstation:~/CPE212_Javillonar$ ls HOA10.1_Log_Monitoring_New
inventory.ini  roles  site.yml
wdjavillonar@workstation:~/CPE212_Javillonar$
```

```
wdjavillonar@workstation:~/CPE212_Javillonar$ git commit -m "Add HOA10.1 Elastic Stack project
"
[main b9f081a] Add HOA10.1 Elastic Stack project
 29 files changed, 537 insertions(+)
 create mode 100644 HOA10.1_Log_Monitoring_New/README.md
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/elasticsearch/README.md
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/elasticsearch/defaults/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/elasticsearch/handlers/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/elasticsearch/meta/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/elasticsearch/tasks/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/elasticsearch/templates/elasticsearch.yml
.j2
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/elasticsearch/tests/inventory
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/elasticsearch/tests/test.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/elasticsearch/vars/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/kibana/README.md
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/kibana/defaults/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/kibana/handlers/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/kibana/meta/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/kibana/tasks/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/kibana/templates/kibana.yml.j2
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/kibana/tests/inventory
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/kibana/tests/test.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/kibana/vars/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/logstash/README.md
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/logstash/defaults/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/logstash/handlers/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/logstash/meta/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/logstash/tasks/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/logstash/templates/logstash.conf.j2
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/logstash/tests/inventory
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/logstash/tests/test.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/roles/logstash/vars/main.yml
 create mode 100644 HOA10.1_Log_Monitoring_New/site.yml
```

```
wdjavillonar@workstation:~/CPE212_Javillonar$ git push origin main
Username for 'https://github.com': MadhelKathlynAnnJavillonar/CPE212_Javillonar
Password for 'https://MadhelKathlynAnnJavillonar%2FCPE212_Javillonar@github.com':
Enumerating objects: 50, done.
Counting objects: 100% (50/50), done.
Delta compression using up to 2 threads
Compressing objects: 100% (26/26), done.
Writing objects: 100% (49/49), 5.83 KiB | 1.46 MiB/s, done.
Total 49 (delta 5), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (5/5), completed with 1 local object.
To https://github.com/MadhelKathlynAnnJavillonar/CPE212_Javillonar.git
   1d8cbaa..b9f081a  main -> main
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.

```
  GNU nano 7.2          roles/elasticsearch/templates/elasticsearch.yml.j2 *
cluster.name: elastic-lab
node.name: "{{ inventory_hostname }}"

path.data: /var/lib/elasticsearch
path.logs: /var/log/elasticsearch

network.host: 0.0.0.0
http.port: 9200
discovery.type: single-node
xpack.security.enabled: false
```